

Proof. From equations (13) and the constraint $\Sigma_1 = \Sigma_2 = \Sigma$, we know that $e_1 = e_2 = e$, where e and τ are unique positive solutions to a pair of fixed point equations. Observe that $K_0 = e\Sigma$ and $K = \gamma\tau K_0 + \lambda I_d = \gamma\tau e \cdot (\Sigma + \theta I_d)$. Defining $\eta := I_{1,1}(\theta)$, one can then rewrite the equations defining e and τ as follows

$$e' = \frac{\lambda}{e} = \lambda + \psi\tau\lambda\bar{\text{tr}}\Sigma K^{-1} = \lambda + \frac{\psi\tau\lambda}{\gamma\tau e}\bar{\text{tr}}\Sigma(\Sigma + \theta I_d)^{-1} = \lambda + \phi\eta e', \quad (162)$$

$$\tau' = \frac{\lambda}{\tau} = \lambda + \lambda\bar{\text{tr}}K_0 K^{-1} = \lambda + \frac{\lambda e}{\gamma\tau e}\bar{\text{tr}}\Sigma(\Sigma + \theta I_d)^{-1} = \lambda + (\eta/\gamma)\tau'. \quad (163)$$

We deduce that

$$e' = \frac{\lambda}{1 - \phi\eta}, \quad \tau' = \frac{\lambda}{1 - \eta/\gamma}, \quad \tau'e' = \lambda\gamma\theta. \quad (164)$$

In particular, the above means that $\eta \leq \min(\gamma, 1/\phi)$. The last part of equations (164) can be rewritten as follows

$$\frac{\lambda}{(1 - \phi\eta)(1 - \eta/\gamma)} = \gamma\theta, \text{ i.e } \phi\eta^2 - (\phi\gamma + 1)\eta + \gamma - \frac{\lambda}{\theta} = 0. \quad (165)$$

This is a quadratic equation for η as a function of λ and θ , with roots

$$\eta^\pm = \frac{\phi\gamma + 1 \pm \sqrt{(\phi\gamma + 1)^2 - 4(\phi\gamma - (\phi/\theta)\lambda)}}{2\phi} = \frac{\psi + 1 \pm \sqrt{(\psi + 1)^2 - 4(\psi - \phi/\theta')}}{2\phi}. \quad (166)$$

Now, for small $\lambda > 0$ and $\psi \neq 1$, we can do a Taylor expansion to get

$$\eta^\pm \simeq \frac{\psi + 1 \pm |\psi - 1|}{2\phi} \pm \frac{1}{\theta|\psi - 1|}\lambda + O(\lambda^2).$$

More explicitly,

$$\begin{aligned} \eta^+ &\simeq O(\lambda^2) + \begin{cases} 1/\phi + \lambda/((1 - \psi)\theta), & \text{if } \psi < 1, \\ \gamma + \lambda/((\psi - 1)\theta), & \text{if } \psi > 1. \end{cases} \\ \eta^- &\simeq O(\lambda^2) + \begin{cases} \gamma - \lambda/((1 - \psi)\theta), & \text{if } \psi < 1, \\ 1/\phi - \lambda/((\psi - 1)\theta), & \text{if } \psi > 1, \end{cases} \end{aligned}$$

Because $\eta \leq \min(1, 1/\phi, \gamma)$, we must have the expansion

$$\begin{aligned} \eta &\simeq O(\lambda^2) + \begin{cases} \gamma - \lambda/((1 - \psi)\theta), & \text{if } \psi < 1, \\ 1/\phi + \lambda/((\psi - 1)\theta), & \text{if } \psi > 1, \end{cases} \\ &= \eta_0 - \frac{1}{(1 - \psi)\theta_0}\lambda + O(\lambda^2), \end{aligned} \quad (167)$$

provided $\theta_0 > 0$, i.e $\eta_0 \neq 1$. in this regime, we obtain

$$\begin{aligned} \tau' &= \frac{\lambda}{1 - \eta/\gamma} \simeq \begin{cases} \lambda/(1 - 1 + \lambda/((1 - \psi)\gamma\theta_0)) = (1 - \psi)\gamma\theta_0, & \text{if } \psi \leq 1, \\ \lambda/(1 - 1/\psi + o(1)) \rightarrow 0, & \text{if } \psi > 1, \end{cases} \\ e' &= \frac{\lambda}{1 - \phi\eta} \simeq \begin{cases} \lambda/(1 - \psi + o(1)) \rightarrow 0, & \text{if } \psi \leq 1, \\ \lambda/(1 - 1 + \lambda\phi/((\psi - 1)\theta_0)) \rightarrow (\psi - 1)\theta_0/\phi, & \text{if } \psi > 1, \end{cases} \\ \tau &= 1 - \eta/\gamma \simeq 1 - \eta_0/\gamma = (1 - 1/\psi)_+, \\ e &= 1 - \phi\eta \simeq 1 - \phi\eta_0 = (1 - \psi)_+. \end{aligned}$$

On the other hand, if $\theta_0 = 0$ (which only happens if $\psi < 1$ and $\gamma > 1$ OR $\psi \geq 1$ and $\phi \leq 1$), it is easy to see from (164) that we must have $\tau' \rightarrow 0$, $e' \rightarrow 0$, $\tau \rightarrow 1 - 1/\gamma$, $e \rightarrow 1 - \phi \geq 0$.

Next, let's compute the limiting values of u and $\omega' := \omega/\tau^2$. □